

Bryan Thomas Carter

Data / ML Engineer & Data Scientist

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I'm a Data Engineer with over **10** years of experience working hands-on with data across the **full pipeline**-from designing and building systems to **optimizing** and **maintaining** them. Over the years, I've worked on projects that involve **machine learning**, **AI**, and **large-scale data analysis**, and I enjoy turning complex data into useful, actionable insights.

I've used tools like **Terraform**, **Docker**, **Kubernetes**, and **MLOps** to build out solid, efficient **CI/CD** workflows and infrastructure setups that follow industry best practices. I've also had the chance to **lead teams**, **mentor junior engineers** and **data scientists**, and help deliver high-quality, real-world results.

I'm someone who genuinely enjoys learning new things and adapting to new tools or environments. I'm always looking for ways to grow and improve, both technically and professionally. What drives me most is solving interesting problems, collaborating with others, and using data to make a meaningful impact.

Skills

- **Language:** Python, SQL, Scala, Shell Scripting, Javascript, PHP
- **Cloud Platforms:** AWS (Redshift, Lambda, S3 etc.), Azure & Google Cloud Platform (BigQuery, DataFlow)
- **Data Engineering:** ETL/ELT, Data Pipelines, Spark, Kafka, Hive, Sqoop, Flume etc.
- **DevOps & Infrastructure:** Kubernetes, Docker, Terraform, Jenkins, GitHub Actions etc.
- **Monitoring & Logging:** CloudWatch, Airflow Metrics, Prometheus, Grafana, Datadog, ELK Stack
- **Databases:** MongoDB, PostgreSQL, Oracle, Cassandra, SQL/NoSQL, Kafka Schema Registry
- **Machine Learning & AI:** MLOps, DVC, Model Deployment, Pandas, Matplotlib, Scikit-Learn, PyTorch
- **Data Modeling:** Dimensional modeling, Star Schema, Snowflake Schema, Data Vault etc.
- **Data Warehousing:** Snowflake, BigQuery, Redshift and Synapse Analytics
- **Visualization Tools:** PowerBI, Tableau, Looker, Seaborn, Matplotlib, Excel
- **Testing:** Great Expectations, Unit Tests for Data Pipelines
- **Streaming:** Apache Flink, Apache Kafka, Amazon Kinesis
- **Familiar with:** Web Scraping, Tensorflow Extended(TFX), MLOps Frameworks, Version Control, Data Governance, Git, Apache Superset etc.

Education

TEXAS TECH UNIVERSITY, LUBBOCK, TX
BACHELOR'S DEGREE IN COMPUTER SCIENCE

SEPTEMBER 2010 - MAY 2014

Experience

DATA ENGINEER LEAD

Kroger | Cincinnati, Ohio | March 2021 – Present

- Engineered scalable **ETL workflows** across **AWS services** such as **Glue**, **EMR** and **S3** to process diverse datasets efficiently.
- Built and **maintained, automated data ingestion systems** with **Python**, integrating structured and unstructured data from multiple systems into **MongoDB**.
- Delivered a highly available **data platform** using **MongoDB** and **Mongo Atlas** on **AWS**, enabling seamless data handling for **healthcare metrics** with full **HIPAA** compliance.
- Optimized MongoDB performance through advanced indexing, **schema design**, and **caching** tailored to analytical dashboard usage.
- Orchestrated unified data engineering and **ML** operations within **Databricks Lakehouse**, streamlining analytics, training, and deployment workflows.
- Constructed star and **snowflake schemas** to enhance warehouse efficiency and **boost SQL query** performance.
- Designed and deployed **Apache, Spark** applications to accelerate batch data processing on **AWS EMR**, improving throughput by 30%.
- Integrated data validation tools into **Spark workflows** to maintain high data integrity.
- Partnered with product and analytics teams to build reliable, scalable pipelines for mission-critical business intelligence.
- Developed **Kafka**-based streaming pipelines to enable near real-time data flows using Spark Streaming.
- Automated infrastructure deployment with **Jenkins, GitLab**, and reusable **Terraform modules**, accelerating delivery cycles.
- Migrated legacy databases to cloud-native solutions such as **RDS** and **Redshift** with **zero downtime** and full data fidelity.
- Applied **AWS Lambda** for lightweight, event-driven transformations, significantly cutting costs and improving responsiveness.
- Implemented **robust CI/CD** pipelines using **Docker** and **GitHub Actions**, ensuring smooth transitions across staging and production environments.
- Reduced manual provisioning efforts by 70% through **Terraform** automation across **AWS** and **Azure** environments.
- Designed high-throughput pipelines capable of processing over **1.3 billion records per hour**, powering real-time analytics.
- Integrated **ML** models into production using **ONNX** and cloud endpoints (**AWS, Vertex AI**) for scalable inference.
- Enhanced system observability and SLA adherence through Prometheus, **Grafana**, and **AWS CloudWatch**.
- Led the build-out of a **hybrid data lake** using **BigQuery** and **GCP Cloud Storage**, improving data discovery and access time.
- Spearheaded the adoption of **Azure Purview** for enterprise-wide data governance and lineage tracking.
- Built machine learning pipelines using **SageMaker**, increasing model prediction accuracy by 25% and shortening iteration cycles.
- Deployed **Redshift**-based analytical systems using **AWS Glue, Lambda** and **S3** for automated data loading.
- Played a key role in cross-functional strategy planning to build reusable analytics assets, resulting in a 15% productivity gain.
- Designed and implemented a **Neo4j**-based knowledge graph to power personalized recommendations with measurable gains.
- Leveraged Great Expectations to automate quality checks, enhancing trust in critical reporting data.
- Guided junior data engineers, fostering a culture of knowledge-sharing and increasing overall team velocity by 22%.
- Streamlined e-commerce analytics infrastructure, slashing query latency and compute costs substantially.
- Developed **MLOps** pipelines to automate model training, deployment, and monitoring for fraud detection in real time.

DATA ENGINEER & MACHINE LEARNING ENGINEER

Deloitte | New York, NY | June 2018 – Feb 2021

- Developed real-time data pipelines using **Apache Kafka, Spark Streaming** and **AWS Kinesis**, capable of processing **6 million events per second**, ensuring high data availability and reducing downtime by 20%.
- Implemented a data lake architecture on **AWS (S3, Glue, and Athena)** to centralize **data storage**, reducing data retrieval latency by 30% and saving a lot of money in infrastructure costs.
- Designed and maintained data warehouse solutions using **Snowflake**, **optimizing query performance by 30%** through clustering and materialized views.

- Designed and implemented **robust authentication** and **authorization mechanisms**, ensuring **secure access control** and **safeguarding sensitive** data across applications.
- Built and maintained continuous integration/continuous delivery (CI/CD) pipelines using **Github Actions**, streamlining the development, testing, and deployment processes.
- Leveraged **AWS** and **Azure** service operators to streamline continuous deployment (CD) from **Github** to **Kubernetes**, enhancing application deployment efficiency.
- Created a distributed **ETL** pipeline using **Airflow** and **Apache Spark** to process 560 GB of data daily, improving job execution times by 35%.
- Increased database connectivity by utilizing **Entity Framework** and **LINQ** to Entities to connect to databases and perform various database transactions, ensuring efficient data management.
- Employed **XSD** and **XSLT** to transform and validate **XML** data, enhancing data interchange and ensuring compatibility across systems for **Data Transformation** and Validation.
- Implemented RESTful APIs and **GraphQL** to communicate with content management systems and data pipelines, enabling data retrieval and manipulation.
- Enhanced testing and reliability by writing automated tests at the unit and environment levels.
- Utilized state-of-the-art **frontend optimization** techniques, including **code splitting**, **lazy loading**, and image optimization, to improve page load times, enhance **user experiences**, and boost search engine rankings.
- Spearheaded the transition from **Angular** to **Blazor** for a suite of enterprise-level web applications, enhancing performance and maintainability while ensuring seamless integration with existing systems.
- Addressed critical memory leaks in the system, optimizing page load times by transitioning from **SASS** to styled-components, effectively managing resource utilization.
- Implemented and customized **MS CRM Dynamics** solutions, including **workflows**, **plugins**, and **custom entities**, to meet specific business needs.
- Analyzed performance by conducting comprehensive unit and integration testing of **Angular** components using **testing frameworks** such as **Jest**, **Jasmine** and **Karma**.
- Analyzed **JMeter** test results to identify bottlenecks and optimize application performance.

MACHINE LEARNING ENGINEER & DATA ANALYST

Texas Instruments | Dallas, Texas | AUGUST 2014 – JAN 2018

- Worked in migration of various back-end **PHP** systems to **C#** services, transforming a significant, monolithic application into a microservices architecture, emphasizing expertise in **.NET** and **Java stack**.
- Demonstrated advanced **DevOps** skills with container orchestrations like **Kubernetes** and managed **CI/CD** operations in **Jenkins** and **Gitlab**, streamlining software deployment in Microsoft ecosystem.
- Employed **Angular** for web-based client-side interfaces and integrated secure real-time communication with **Django** and **WebSocket** technologies on the backend, illustrating full-stack expertise.
- Integrated **Elasticsearch**, **Postgres**, and **PostGIS** to develop a location-based search engine.
- Developed REST APIs using **Python**, **Django**, **Rest Framework** and **MongoDB** for **robust** data management.
- Configured **uWSGI** to handle high connection loads, enabling real-time communication between a **Python** web server and the **JavaScript-based frontend** which aligns with full-stack development.
- Utilized **Scrapy** to extract data from social platforms, showcasing data handling skills within the **Microsoft stack**.
- Enhanced the frontend development process, including coding guidelines and code quality tools, promoting best practices for **JavaScript** in **frontend** development within the Microsoft technology stack.
- Employed Node.js and **Redux** to interact with **DatoCMS** for Headless **CMS**, emphasizing full-stack development with Microsoft technologies for API integration.
- Designed, built and conducted unit testing of software in a collaborative environment, showcasing teamwork and software development skills within the Microsoft technology ecosystem.